

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5101

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I S. Modhuri (192465031) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

S. Modhuri
Signature of the student:

98
100
A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.
 - a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
 - b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.
2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
 - a. Deduce the most likely original message.
 - b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.
3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.
 - a. Demonstrate how the encryption process transforms the message step-by-step.
 - b. Explain how the repeating keyword influences each character of the ciphertext.
4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.
 - a. Analyze how investigators could detect the presence of hidden data within the image.
 - b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.
5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEAOCAIVDEN".
 - a. Determine the most probable original message by reversing the encryption process.
 - b. Critically evaluate why such a method is vulnerable in modern communication systems.
 - c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 23	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

98
100
100

Event communication security:

Aim:

To encrypt & decrypt the confidential message "MEET AT NOON" using the caesar cipher with a shift value of 4 & demonstrate secure communication using a classical encryption technique

Theory:

Caesar Cipher is a monoalphabetic substitution cipher in which each letter of the plaintext is replaced by another letter located a fixed number of positions away in the alphabet.

→ A shift value (key) is used for encryption & decryption.

Encryption: $C = (P + k) \bmod 26$

where:

- C = ciphertext
- P = plaintext
- k = Shift value
- $\bmod 26$ = total alphabets.

Decryption:

$$P = (C - k) \bmod 26$$

Input:

Plaintext = MEET AT NOON

Key = 4.

Encryption

Plain text	Position	$(P+K) \bmod 26 = C$	Ciphertext letter
M	12	$(12+4) \bmod 26 = 16$	Q
E	4	$(4+4) \bmod 26 = 8$	I
E	4	$(4+4) \bmod 26 = 8$	I
T	19	$(19+4) \bmod 26 = 23$	X
A	0	$(0+4) \bmod 26 = 04$	E
T	19	$(19+4) \bmod 26 = 23$	X
N	13	$(13+4) \bmod 26 = 17$	R
O	14	$(14+4) \bmod 26 = 18$	S
O	14	$(14+4) \bmod 26 = 18$	S
N	13	$(13+4) \bmod 26 = 17$	R

Ciphertext generated:

MEET AT NOON $\rightarrow$ Q I I X E X R S S R

Decryption process:

The authorized member who knows the shift value can recover the original message.

Cipher letter	position	$P = (C - K) \bmod 26$	Plain letter
Q	16	$(16-4) \bmod 26 = 12$	M
I	8	$(8-4) \bmod 26 = 4$	E
I	8	$(8-4) \bmod 26 = 4$	E
X	23	$(23-4) \bmod 26 = 19$	T
E	4	$(4-4) \bmod 26 = 0$	A
X	23	$(23-4) \bmod 26 = 19$	T
R	17	$(17-4) \bmod 26 = 13$	N
S	18	$(18-4) \bmod 26 = 14$	O
S	18	$(18-4) \bmod 26 = 14$	O
R	17	$(17-4) \bmod 26 = 13$	N

Plaintext generated:

QIIX EX ASSR → MEET AT NOON

Result analysis:

- The original message was successfully transformed into ciphertext.
- Unauthorized users cannot directly understand the message.
- The receiver can recover the original message using the same key.
- This demonstrates confidentiality in communication.

Advantages:

- Easy to implement.
- Fast encryption & decryption.
- Useful for learning cryptographic principles.

Limitations:

- Only 25 possible keys.
- Easily broken using brute-force attacks.
- Not suitable for modern secure systems.

Conclusion:

The message "MEET AT NOON" was encrypted as "QIIX EX ASSR" using Caesar cipher with a shift of 4. The original message was successfully recovered using the decryption process, demonstrating the basic principle of secure communication.

3. Secure Messaging System Using Vigenere Cipher

Aim:

To encrypt the plaintext HELLO using the keyword KEY & analyze the effect of a repeating keyword in the Vigenere Cipher.

Theory:

Vigenere Cipher is a polyalphabetic substitution cipher that uses a keyword to generate different shift values for different characters.

• Unlike Caesar Cipher, the shift changes for every letter based on the keyword.

Input:

Plaintext = HELLO

Keyword = KEY

Keyword Expansion:

Since the plaintext contains five characters, the keyword repeated:

Plaintext	H	E	L	L	O
Keyword	K	E	Y	K	E
P value	7	4	11	11	14
K value	10	4	24	10	4

Alphabetical Values:

A	B	C	D	E	F	G	H	I	J	K	L	M	N
0	1	2	3	4	5	6	7	8	9	10	11	12	13
O	P	Q	R	S	T	U	V	W	X	Y	Z		
14	15	16	17	18	19	20	21	22	23	24	25		

Encryption:

$$C = (P + K) \bmod 26$$

Encryption process:

Plain letter	P	Key letter	K	Calculation	Cipher letter
H	7	K	10	$7 + 10 = 17$	R
E	4	E	4	$4 + 4 = 8$	I
L	11	Y	24	$35 = 9$	J
L	11	K	10	21	V
O	14	E	4	18	S

Ciphertext:

HELLO $\rightarrow$ RIJVS

Decryption Formula:

$$P = (C - K) \bmod 26$$

Decryption process:

Cipher letter	C	Key letter	K	Calculation	plain letter
R	17	K	10	$17 - 10 = 7$	H
I	8	E	4	$8 - 4 = 4$	E
J	9	Y	24	$24 - 9 = 15$ $9 - 24 = -15 \bmod 26$ $= 26 - 15 = 11$	L
V	21	K	10	$21 - 10 = 11$	L
S	18	E	4	$18 - 4 = 14$	O

Plain text:

RIJVS $\rightarrow$ HELLO

Influence of Repeating keyword:

- The keyword KEY is repeated as KEYKE
- Different shift values are applied to different letters
- The same plaintext letter may produce different ciphertext letters.
- Frequency analysis becomes difficult
- Security is stronger than caesar cipher

Result analysis:

- The plaintext HELLO was successfully encrypted as RIJVS
- The keyword controlled the shift applied to each character
- The varying shifts improved confidentiality
- The encryption process reduced predictable patterns in the ciphertext.

Advantages:

- More secure than monoalphabetic substitution ciphers.
- Uses multiple shifting values.
- Difficult to crack using simple frequency analysis
- Provides better confidentiality.

Limitations:

- Short keywords reduce security
- Repeated keyword may reveal patterns
- Not as secure as modern algorithms like AES.

Conclusion:

The Vigenere Cipher successfully encrypted HELLO into RIJVS using the keyword KEY. The use of a repeating keyword generated multiple shift values, making the cipher stronger than a simple Caesar cipher & improving message confidentiality.